

JEGADEESH D

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PROFESSIONAL SUMMARY

Software Engineer with an Integrated M.Tech in Software Engineering (Sept 2025). Experienced in building production-grade agent runtimes, hybrid RAG pipelines, and high-throughput ML inference microservices using Python, FastAPI, TypeScript, React 18, and PostgreSQL/SQLite. Focused on deterministic API contracts, test-driven engineering (239+ automated pytest tests across personal systems), quantitative LLM evaluation calibration, and containerized Docker deployments.

CORE COMPETENCIES & TECHNICAL SKILLS

AI & Agentic Systems — Agent Runtimes, Tool-Calling, Hybrid RAG (Dense + BM25), Reciprocal Rank Fusion (RRF), ChromaDB, Sentence Transformers, LLM APIs (OpenRouter, Llama-3.3), scikit-learn, PyTorch

Backend & Systems — Python (AsyncIO), FastAPI, REST APIs, Docker, PostgreSQL, SQLite (FTS5), Redis, Linux, Git, Uvicorn, Pydantic-Settings

Testing & Reliability — pytest (239+ Automated Tests), pytest-asyncio, Evaluator Calibration Benchmarks, Unit & Integration Testing, CI/CD

Full-Stack & Frontend — TypeScript, React 18, Vite, Schema Validation, Responsive UI, Streamlit (Internal Ops Cockpits)

FLAGSHIP SOFTWARE SYSTEMS

Financial Regulatory Intelligence Engine — Hybrid Dense + Sparse RAG *Python, FastAPI, ChromaDB, BM25, Docker, pytest*

- Architected an enterprise hybrid retrieval pipeline over 3,400+ BFSI regulatory filings (RBI, SEBI, HDFC), fusing dense MiniLM embeddings with BM25 sparse keyword ranking via Reciprocal Rank Fusion (RRF; $k=60$).
- Benchmarked retrieval offline across 20 regulatory test suites, lifting Mean Reciprocal Rank (MRR) to 0.550 and Top-5 Hit Rate to 70.0% independently of LLM generation.
- Deployed containerized FastAPI service with Portkey gateway failover, confidence gating, and page-level citations; validated by 83 automated pytest tests. [\[GitHub\]](#)

GoalOS — Local-First Executive Agent Runtime *React 18, TypeScript, FastAPI, SQLite FTS5, ChromaDB*

- Engineered a local-first agent runtime with React 18 + Vite frontend and FastAPI backend, achieving 100% tool-calling accuracy (7/7) across multi-step benchmark tasks.
- Implemented a 5-factor composite memory retrieval engine combining SQLite FTS5 full-text lexical search with ChromaDB cosine similarity, exponential recency decay ($e^{-\lambda \Delta t}$), and importance weighting for zero-latency recall.
- Built automatic failover to deterministic local heuristic engines during offline operation or API rate limits; validated by 109 automated pytest tests. [\[GitHub\]](#)

SuperKalam — Agentic UPSC Mains Evaluator *Python, FastAPI, SQLite, ChromaDB, OpenRouter, Pydantic v2*

- Designed an autonomous multi-agent pipeline (Retrieval → Evaluator → Feedback) scoring student answers against official Civil Services rubrics across 4 dimensions with native Indic script feedback in Hindi and Tamil.
- Calibrated evaluator agent scoring across 40 Previous Year Questions over 120 benchmark runs, achieving Pearson $r = 0.9551$, Spearman $\rho = 0.9736$, and MAE = 0.58 against ground-truth performance tiers.
- Enforced Pydantic v2 JSON contracts with automated retry handling and HTTP 429 circuit breaking; validated by 25 automated pytest tests. [\[GitHub\]](#)

Customer Support Analytics — Multi-Task ML & Agentic Triage *Python, FastAPI, scikit-learn, PostgreSQL, Docker*

- Architected a two-tier triage system over ~200K tickets: Tier 1 classical models resolve 90% of routine traffic at sub-15ms latency (\$0 API cost; 82.1% priority accuracy, $R^2 = 0.719$ resolution regression with zero data leakage).
- Implemented Tier 2 agentic escalation activating only for low confidence (<0.80) or severe resolution ($>48h$), generating structured JSON with customer frustration diagnostics and auto-drafted replies.
- Deployed FastAPI REST endpoints (POST /predict_*, POST /triage_agent) with PostgreSQL orchestration; validated by 15 automated pytest tests. [\[GitHub\]](#)

EDUCATION

MTech (Integrated) in Software Engineering — Vellore Institute of Technology (VIT), Chennai *Jul 2020 – Sep 2025*

CGPA: 7.89 / 10.0 | **Coursework:** Data Structures & Algorithms, Software Architecture, Distributed Systems, Database Systems, Machine Learning

LEADERSHIP & EXTRACURRICULAR

Photography Head (Operations Lead) — Vibrance & TechnoVIT (International Fests), VIT Chennai *Jan 2024*

- Directed a 70-member operations crew for multi-day international fests, driving live execution, asset tracking, and strict schedule adherence.
- Standardized event operational workflows and asset delivery SOPs, coordinating across 15+ university committees and vendor teams.